Rational Functions and Asymptotes

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Finding asymptotes and intercepts

For $f(x) = \frac{x+2}{x-3}$, find:

a the vertical asymptote

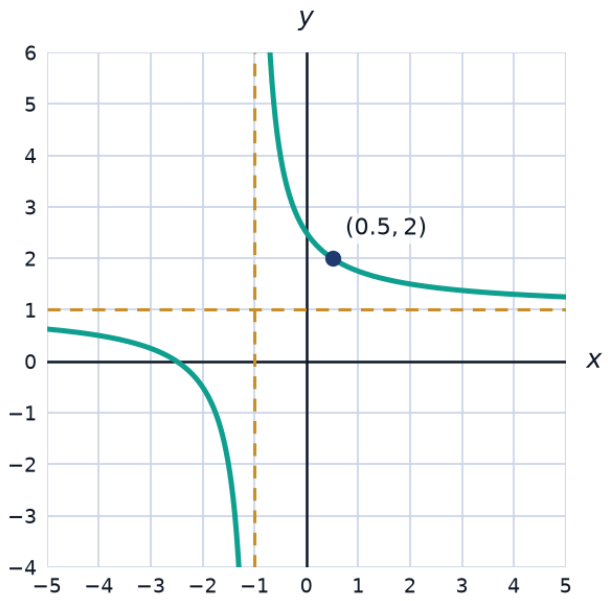
b the horizontal asymptote

c the x-intercept

d the y-intercept

2 [Foundational] • Target: Finding asymptotes and intercepts

Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.



3 [Foundational] • Target: Finding asymptotes and intercepts

Consider $f(x) = \frac{x^2-9}{x-3}$.

a Simplify f and identify the feature at $x = 3$.

b Does the graph have a vertical asymptote?

4 [Foundational] • Target: Finding asymptotes and intercepts

Find the slant (oblique) asymptote of $f(x) = \frac{x^2+1}{x-1}$.

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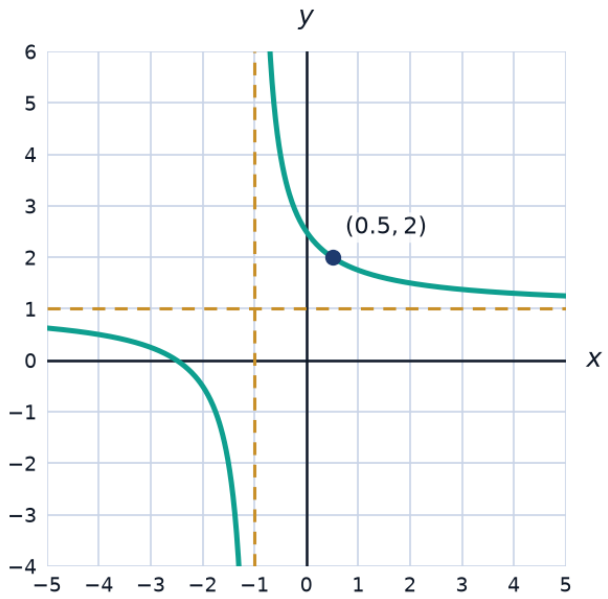
5 [Foundational] • Target: Behaviour near asymptotes

For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:

a $x \to 3^+$ **b** $x \to 3^-$ **c** $x \to \infty$

6 [Foundational] • Target: Behaviour near asymptotes

Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.



7 [Foundational] • Target: Classifying end behaviour from degree

Without dividing, state the horizontal or slant asymptote (or neither) for each function based on the degrees of numerator and denominator:

a $f(x) = \frac{3x+1}{x^2-4}$ (degree 1 over 2)

b $f(x) = \frac{5x^2-1}{2x^2+3}$ (degree 2 over 2)

c $f(x) = \frac{x^3+2}{x^2-1}$ (degree 3 over 2)

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Classifying end behaviour from degree

Find the slant asymptote of $f(x) = \frac{2x^2-3x+1}{x+2}$.

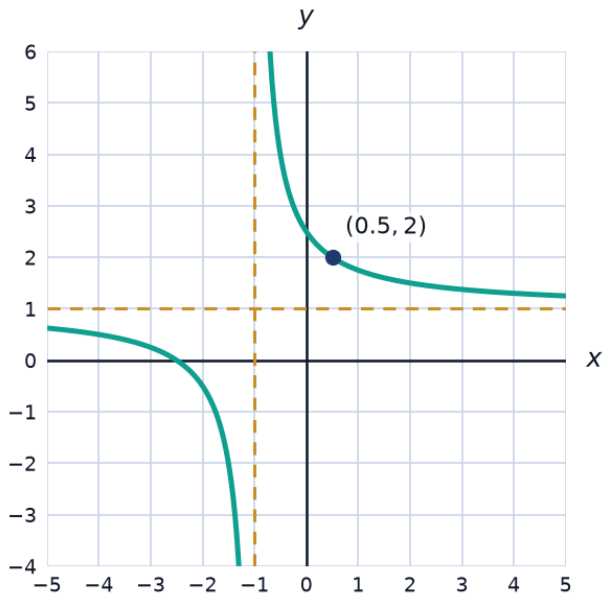
9 [Intermediate] • Target: Holes versus asymptotes

For $f(x) = \frac{(x-2)(x+5)}{(x-2)(x-1)}$, identify the hole and the vertical asymptote.

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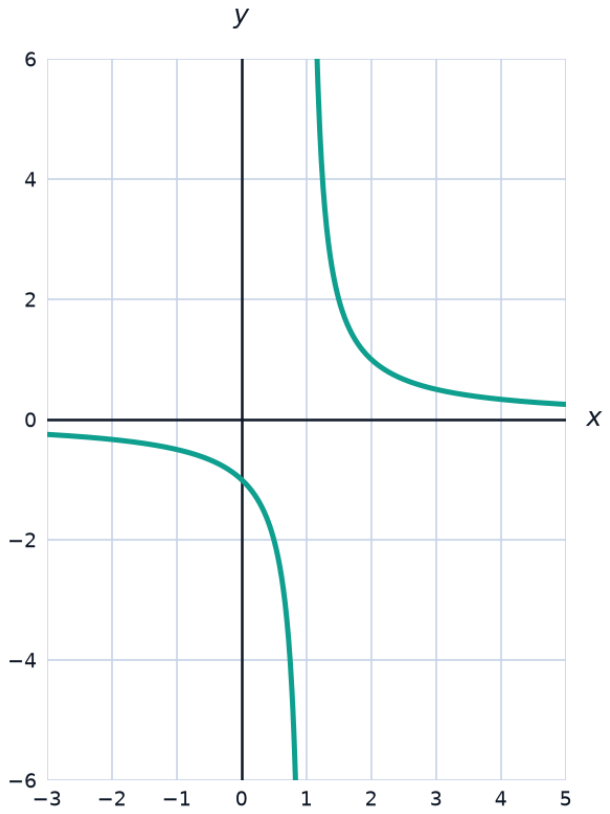
10 [Intermediate] • Target: Holes versus asymptotes

Construct a rational function that has a hole at $x = 3$ and a vertical asymptote at $x = -1$.



11 [Intermediate] • Target: Reading asymptotic behavior from a graph

The graph of $f(x) = \frac{1}{x-1}$ is shown. Use it to describe the behavior near $x = 1$ and as $x \to \pm\infty$.



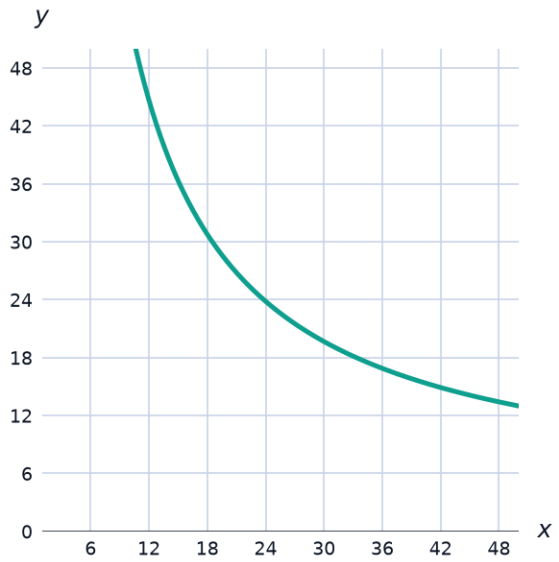
12 [Intermediate] • Target: Domain restrictions and sign analysis

Find the domain of $f(x) = \frac{x-1}{x^2-x-6}$, and determine the sign of $f(x)$ on the interval $(3, \infty)$.

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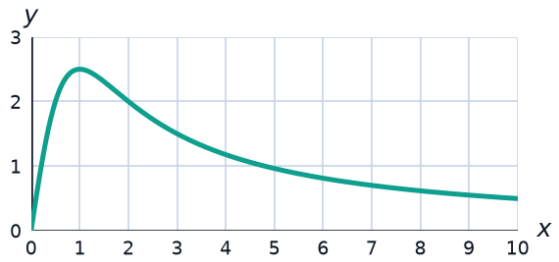
13 [Intermediate] • Target: Asymptotes in applied contexts

The average cost per unit of producing x items is $C(x) = \frac{500 + 3x}{x}$ (setup cost 500 riyals plus 3 riyals per unit). Find the horizontal asymptote and explain its meaning in context.



14 [Intermediate] • Target: Asymptotes in applied contexts

A drug's concentration in the bloodstream is modeled by $C(t) = \frac{5t}{t^2 + 1}$ (mg/L, hours). Find the horizontal asymptote and interpret it, and find $C(0)$.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

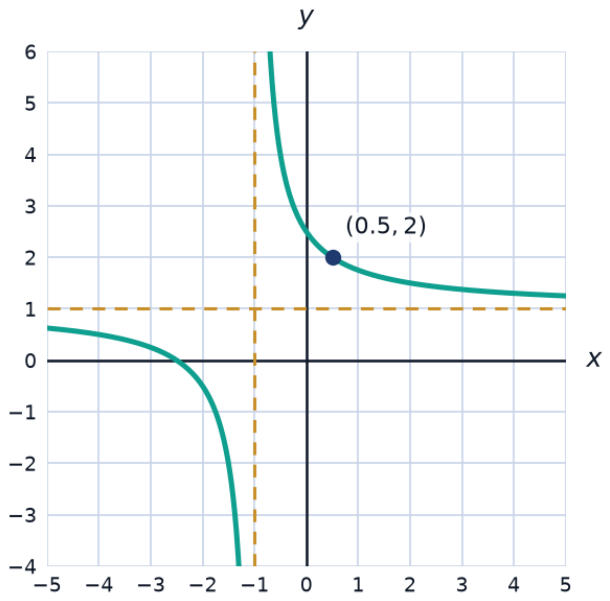
15 [Challenging] • Target: Finding a rational function's equation from its features • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the equation of a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x-intercept at $x = 1$.

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16 [Challenging] • Target: Holes versus asymptotes • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$? Show your complete algebraic work.



17 [Challenging] • Target: Reading asymptotic behavior from a graph • Cross-Lesson: 4.01 Circles and Parabolas

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

18 [Challenging] • Target: Domain restrictions and sign analysis • Cross-Lesson: 4.02 Ellipses

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 5}{x - 1}$.

19 [Challenging] • Target: Asymptotes in applied contexts • Cross-Lesson: 4.03 Hyperbolas

Find the slant (oblique) asymptote of $f(x) = \frac{3x^2 - 12x + 12}{x + 5}$.

20 [Challenging] • Target: Finding a rational function's equation from its features • Cross-Lesson: 5.01 Vectors: Components and Operations

Find the slant (oblique) asymptote of $f(x) = \frac{2x^2 + 12x + 18}{x - 5}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vertical Asymptotes and Holes

Find all vertical asymptotes and holes for $f(x) = \frac{x^2 - 4}{x^2 - x - 2}$.

a. Hole at $x = 2$, Vertical asymptote at $x = -1$

b. Vertical asymptotes at $x = 2$ and $x = -1$

c. Hole at $x = -1$, Vertical asymptote at $x = 2$

d. No holes, vertical asymptote at $x = 2$

22 [Foundational] • Target: Vertical Asymptotes and Holes

Find the coordinates (x, y) of the hole for $f(x) = \frac{x^2 - 9}{x - 3}$.

a. $(3, 6)$

b. $(3, 0)$

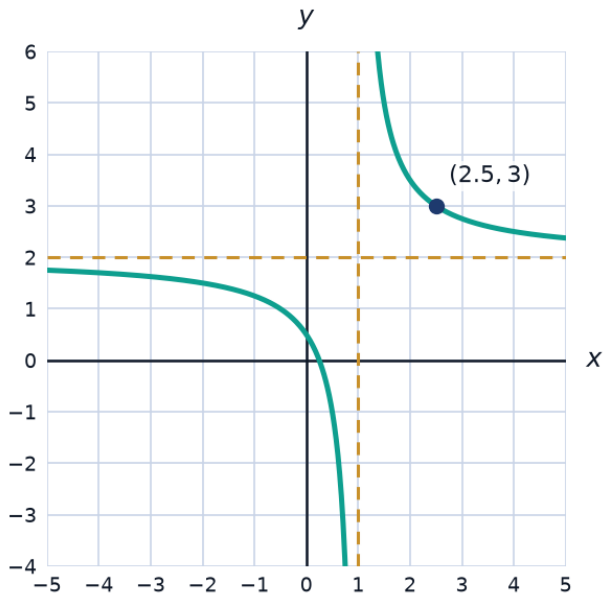
c. $(3, 3)$

d. $(-3, 0)$

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23 [Foundational] • Target: Vertical Asymptotes and Holes

How do you determine if a real zero of the denominator causes a hole or a vertical asymptote?



a. If the numerator also has the same zero of equal or greater multiplicity, it is a hole; otherwise it is a vertical asymptote

b. Every zero of the denominator is an asymptote

c. If the degree of the numerator is larger, it is always a hole

d. Check the sign of $f(0)$

24 [Foundational] • Target: Horizontal Asymptotes

Find the horizontal asymptote of $f(x) = \frac{4x^2 + 1}{3x^2 - 5x + 2}$.

a. $y = 4/3$

b. $y = 0$

c. No horizontal asymptote

d. $y = 1/2$

25 [Foundational] • Target: Horizontal Asymptotes

Find the horizontal asymptote of $g(x) = \frac{7x - 2}{x^3 + 4}$.

a. $y = 0$

b. $y = 7$

c. No horizontal asymptote

d. $y = -1/2$

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26 [Foundational] • Target: Horizontal Asymptotes

Under what condition does a rational function have NO horizontal asymptote?

a. When the degree of the numerator is strictly greater than the degree of the denominator

b. When the degree of the denominator is greater

c. When the degrees are equal

d. When there are no real roots

27 [Foundational] • Target: Slant (Oblique) Asymptotes

When does a rational function $f(x) = \frac{P(x)}{Q(x)}$ possess a slant (oblique) asymptote?

a. When the degree of the numerator is exactly 1 greater than the degree of the denominator

b. When the degree of the numerator is 2 greater

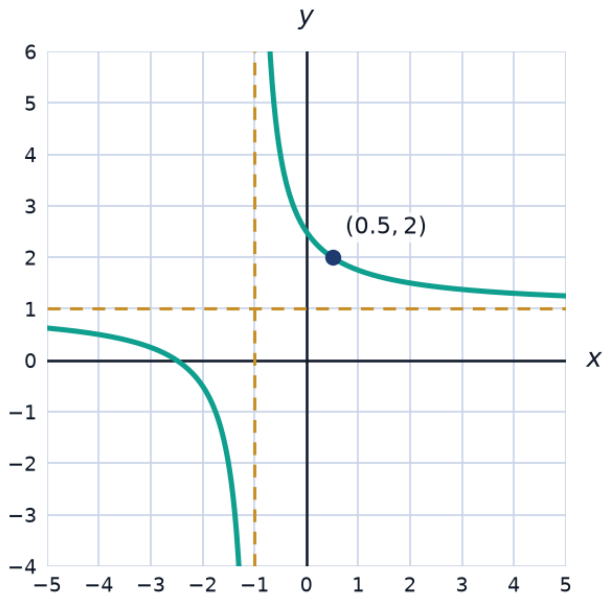
c. When the degree of the denominator is greater

d. When there are no vertical asymptotes

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28 [Foundational] • Target: Slant (Oblique) Asymptotes

Find the slant asymptote of $f(x) = \frac{2x^2 + 7x + 5}{x + 2}$.



a. $y = 2x + 3$

b. $y = 2x + 7$

c. $y = 2x - 1$

d. $y = x + 3$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Slant (Oblique) Asymptotes

Find the slant asymptote of $g(x) = \frac{x^3 - 1}{x^2 + 1}$.

a. $y = x$

b. $y = x - 1$

c. $y = 1$

d. No slant asymptote

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30 [Intermediate] • Target: Intercepts and Asymptote Crossings

Find all x-intercepts and the y-intercept of $f(x) = \frac{x^2 - 1}{x^2 - 4}$.

a. x-intercepts: $(\pm 1, 0)$, y-intercept: $(0, 1/4)$

b. x-intercepts: $(\pm 2, 0)$, y-intercept: $(0, -1/4)$

c. x-intercepts: $(\pm 1, 0)$, y-intercept: $(0, -1)$

d. x-intercepts: $(1, 0)$, y-intercept: $(0, 1/4)$

31 [Intermediate] • Target: Intercepts and Asymptote Crossings

Can a rational function cross its horizontal or slant asymptote?

a. Yes, a function can cross its horizontal or slant asymptote at finite values of x

b. No, a graph can never touch any asymptote

c. Only at the y-intercept

d. Only if it has no vertical asymptotes

32 [Intermediate] • Target: Intercepts and Asymptote Crossings

Find the point where $f(x) = \frac{x^2 + 3x - 4}{x^2 - 1}$ crosses its horizontal asymptote $y = 1$.

a. $(-1, 1)$ is undefined; crossing is at $x = 1$ is undefined; crosses at $(1, 1)$ is a hole; no crossing point

b. It crosses at $x = 1$

c. It crosses at $x = 0$

d. It does not cross its horizontal asymptote

33 [Intermediate] • Target: Finding asymptotes and intercepts

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 1}{x - 1}$.

a. $y = 6x$

b. $y = x + 6$

c. $y = x + 7$

d. No slant asymptote

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34 [Intermediate] • Target: Behaviour near asymptotes

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 4x + 1}{x - 1}$.

a. $y = 5x$

b. No slant asymptote

c. $y = x + 5$

d. $y = x + 6$

35 [Intermediate] • Target: Classifying end behaviour from degree

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x - 4}{x - 1}$.

a. $y = x + 7$

b. $y = x + 6$

c. $y = 6x$

d. No slant asymptote

36 [Intermediate] • Target: Holes versus asymptotes

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 2x - 1}{x - 1}$.

a. $y = x + 4$

b. $y = 3x$

c. $y = x + 3$

d. No slant asymptote

37 [Intermediate] • Target: Reading asymptotic behavior from a graph

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 2x + 3}{x - 1}$.

a. $y = x + 3$

b. $y = 3x$

c. $y = x + 4$

d. No slant asymptote

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Domain restrictions and sign analysis • Cross-Lesson: 4.02 Ellipses

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 2x + 5}{x - 1}$.

a. $y = 3x$

b. $y = x + 4$

c. No slant asymptote

d. $y = x + 3$

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39 [Challenging] • Target: Asymptotes in applied contexts • Cross-Lesson: 4.03 Hyperbolas

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 3}{x - 1}$.

a. No slant asymptote

b. $y = 6x$

c. $y = x + 6$

d. $y = x + 7$

40 [Challenging] • Target: Finding a rational function's equation from its features • Cross-Lesson: 5.01 Vectors: Components and Operations

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x - 2}{x - 1}$.

a. $y = x + 7$

b. $y = 6x$

c. $y = x + 6$

d. No slant asymptote

41 [Challenging] • Target: Vertical Asymptotes and Holes • Cross-Lesson: 5.02 The Dot Product

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 2}{x - 1}$.

a. $y = x + 7$

b. $y = 6x$

c. $y = x + 6$

d. No slant asymptote

42 [Challenging] • Target: Horizontal Asymptotes • Cross-Lesson: 5.03 Parametric Equations

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x - 5}{x - 1}$.

a. No slant asymptote

b. $y = x + 6$

c. $y = x + 7$

d. $y = 6x$

43 [Challenging] • Target: Slant (Oblique) Asymptotes • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 4x + 3}{x - 1}$.

a. $y = x + 6$

b. No slant asymptote

c. $y = 5x$

d. $y = x + 5$

44 [Challenging] • Target: Intercepts and Asymptote Crossings • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 4x - 2}{x - 1}$.

a. $y = x + 5$

b. No slant asymptote

c. $y = 5x$

d. $y = x + 6$

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45 [Challenging] • Target: Finding asymptotes and intercepts • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 2x - 5}{x - 1}$.

a. $y = 3x$

b. No slant asymptote

c. $y = x + 4$

d. $y = x + 3$

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